Recombinant expression by IPTG induction

What you're going to get / What you expect

Overexpression of the target enzyme in E. coli, starting from a colony on the selective plate and inducing the T7 system with IPTG. Induction is done at a reduced temperature to favour native folding.

Materials list (per 10 mL production culture)

- Selective plate with colonies of the transformed (DE3) expression strain.
- LB medium with the relevant antibiotic.
- IPTG (isopropyl-b-D-1-thiogalactopyranoside), 1 M stock.
- Orbital shaker; spectrophotometer (OD600).

Steps

1. Pick a single colony from the selective plate into 5 mL of LB with antibiotic and grow overnight at 37 ºC under orbital agitation (pre-inoculum).
2. Inoculate the 10 mL production culture in LB with antibiotic to an initial OD600 of 0.05.
3. Grow at 37 ºC to mid-exponential phase (OD600 between 0.4 and 0.6).
4. Induce by adding IPTG to a final concentration of 1 mM.
5. Incubate overnight at a reduced 20 ºC to slow protein synthesis and promote native folding.
6. Collect biomass for the expression test.

Errores comunes / troubleshooting

- Strong bands in the total fraction but nothing soluble: the limitation is folding, not yield; move to solubilisation strategies (induction at 30 C, co-expression of chaperones, oxidising or rare-codon strains).
- A sharp drop in OD600 shortly after induction points to a cytotoxic enzyme (as for phospholipases C): do not force it in a live host, reroute to cell-free in vitro transcription and translation (IVTT).